4：Biostatistics
一般化線形モデル

Logistic, Poisson

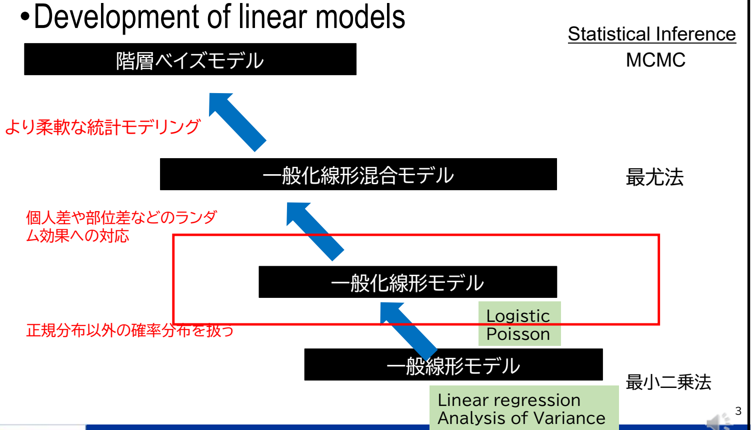
Course Content

This course will be divided in 11 chapters as follow

Review

一般線形モデル
一般化線形モデル

- Development of linear models



Review

一般線形モデル?

- 一般線形モデル

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_i x_i + \varepsilon$$

$$\boxed{y = X\beta + e}$$ It can be expressed by this equation

- 重回帰分析(i ≤ 2)

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{n1} & x_{n2} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} \qquad \boxed{y = X\beta + e}$$

- ANOVA, ANCOVA

$$\begin{bmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \\ y_{31} \\ y_{32} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu \\ T_1 \\ T_2 \\ T_3 \end{bmatrix} + \begin{bmatrix} e_{11} \\ e_{12} \\ e_{21} \\ e_{22} \\ e_{31} \\ e_{32} \end{bmatrix} \qquad \boxed{y = X\beta + e}$$

Review

GLMは以下の3つの要素で構成される

1. 確率分布の指数族
   指数族(二項、ガンマ、ポアソンなど)

$$\eta = X\beta$$

2. 線形予測量 η
線形予測量( )は独立変数の情報をモデルに取り込む量である。
ηは未知パラメータβの線形結合(したがって「線形」)で表され、線形結合の係数は独立変数の行列Xとして表される。

3. リンク関数
ある式を線形に変換する関数。

リンク関数

- 式を線形に変換する関数

	Probability Distribution	glm() の family	Frequently used link functions	Variance
Discrete variable	Bernoulli	binomial	logit	$\mu(1-\mu)$
	Binomial	binomial	logit	$\mu(1-\mu)$ (注)
	Poisson	poisson	log	μ
Continuous variable	Gamma	gamma	log かな?	μ^2
	Normal	gaussian	identity	一定

アウトカム変数(種類と相関)および解析

Correlation of outcome variables

outcome variables	(−)	(+)
continuous	General linear model (GLM) ・regression analysis(REG) ・analysis of variance (ANOVA) ・t-test (TTEST)	General linear mixed model (MIXED)
binominal	Logistic regression (LOGISTIC)	Generalized linear mixed model (NLMIXED) (GLIMMIX)
multiple	Loglinear model (CATMOD)	
Count data	Poisson regression (GENMOD)	
	Generalized linear model (GENMOD)	

Logistic model

- 二項ロジスティック回帰は, バイナリ応答変数が一連の説明変数に関係する回帰の特殊なタイプである.
- ロジスティック回帰では, 応答が特定の値をとる確率またはオッズが, 予測変数によってとられる値の組み合わせに基づいてモデル化される.

Example
データセット 神経痛(N=60)

高齢者における神経痛の鎮痛剤に関する研究

反応変数 痛みの消失率(痛みの有無:有、無)
グループ変数 治療法(Treatment: P(プラセボ)、A、B)
共変量
Sex(性別:F、M)
年齢(治療開始時の年齢:連続変数)
Duration(罹病期間:連続量(月)

Data set (Data=Neuralgia)

```
Data Neuralgia;
input Treatment $ Sex $ Age Duration Pain $ @@;
datalines;
P F 68 1 No B M 74 16 No P F 67 30 No P M 66 26 Yes
B F 67 28 No B F 77 16 No A F 71 12 No B F 72 50 No
B F 76 9 Yes A M 71 17 Yes A F 63 27 No A F 69 18 Yes
B F 66 12 No A M 62 42 No P F 64 1 Yes A F 64 17 No
P M 74 4 No A F 72 25 No P M 70 1 Yes B M 66 19 No
B M 59 29 No A F 64 30 No A M 70 28 No A M 69 1 No
B F 78 1 No P M 83 1 Yes B F 69 42 No B M 75 30 Yes
P M 77 29 Yes P F 79 20 Yes A M 70 12 No A F 69 12 No
B F 65 14 No B M 70 1 No B M 67 23 No A M 76 25 Yes
P M 78 12 Yes B M 77 1 Yes B F 69 24 No P M 66 4 Yes
P F 65 29 No P M 60 26 Yes A M 78 15 Yes B M 75 21 Yes
A F 67 11 No P F 72 27 No P F 70 13 Yes A M 75 6 Yes
B F 65 7 No P F 68 27 Yes P M 68 11 Yes P M 67 17 Yes
B M 70 22 No A M 65 15 No P F 67 1 Yes A M 67 10 No
P F 72 11 Yes A F 74 1 No B M 80 21 Yes A F 69 3 No
run;
```

The code and dataset are stored on the web class.

Additional explanation of the code.
"@@ "is an instruction to read repeatedly in the order of variables before "@@".

The data is entered continuously on a single line, but it is read in five variables at a time: "Treatment," "Sex," "Age," "Duration," and "Pain.

The same as
P F 68 1 No B M 74 16 No
P F 67 30 No P M 66 26 Yes
B F 67 28 No B F 77 16 No
A F 71 12 No B F 72 50 No
・
・

Logistic regression

ロジスティック回帰では、従属変数はオッズの自然対数であるロジットである、

$$\log(odds) = logit(P) = \ln\left(\frac{p}{1-p}\right)$$

つまり、ロジットとはオッズの対数であり、オッズはPの関数である。 ロジスティック回帰では、次のようになる。

$$logit(P) = \ln\left(\frac{p}{1-p}\right) = a + bX$$

$$\frac{p}{1-p} = e^{a+bx}$$

$$p = \frac{e^{a+bx}}{1+e^{a+bx}}$$

LOGISTIC SAS program

```
ods graphics on;
proc logistic PLOTS=(ODDSRATIO EFFECT)
data=neuralgia;
class treatment sex/param=glm;
model pain= treatment;
oddsratio treatment;
run;
ods graphics off;
```

Study of Analgesics for Neuralgia in the Elderly PROC logistic

```
ods graphics on;
proc logistic data=Neuralgia
PLOTS=(ODDSRATIO(TYPE=HORIZONTALSTAT));
class treatment sex / param=glm;
model pain= treatment sex age duration;
                                    [**units duration=5]
lsmeans treatment/adj=tukey cl exp;
run;
```

Comparison with reference adj=Dunnett
Comparison between all group adj=Tukey

Output of Result PROC LOGISTIC

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	15.5744	6.5915	5.5828	0.0181
Treatment	A	1	3.1817	1.0161	9.8049	0.0017
Treatment	B	1	3.7085	1.1407	10.5700	0.0011
Treatment	P	0	0	.	.	.
Sex	F	1	1.8322	0.7963	5.2946	0.0214
Sex	M	0	0	.	.	.
Age		1	-0.2621	0.0970	7.2977	0.0069
Duration		1	0.00586	0.0330	0.0315	0.8591

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Treatment A vs P	24.087	3.288	176.481
Treatment B vs P	40.794	4.362	381.552
Sex F vs M	6.248	1.312	29.750
Age	0.769	0.636	0.931
Duration	1.006	0.943	1.073

Multiple comparison

$$\ln\left(\frac{p}{1-p}\right) = a + bX$$

$$\frac{p}{1-p} = e^{a+bX}$$

The LOGISTIC Procedure

Differences of Treatment Least Squares Means Adjustment for Multiple Comparisons: Tukey-Kramer												
Treatment	_Treatment	Estimate	Standard Error	z Value	Pr > |z|	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper	Exponentiated
A	B	-0.5269	0.9371	-0.56	0.5740	0.8402	0.05	-2.3635	1.3098	-2.7231	1.6694	0.5905
A	P	3.1817	1.0161	3.13	0.0017	0.0050	0.05	1.1902	5.1732	0.8003	5.5631	24.0874
B	P	3.7085	1.1407	3.25	0.0011	0.0033	0.05	1.4728	5.9442	1.0351	6.3820	40.7942

Differences of Treatment Least Squares Means Adjustment for Multiple Comparisons: Tukey-Kramer					
Treatment	_Treatment	Exponentiated Lower	Exponentiated Upper	Adj Lower Exp	Adj Upper Exp
A	B	0.09409	3.7053	0.06567	5.3088
A	P	3.2876	176.48	2.2261	260.64
B	P	4.3616	381.55	2.8154	591.09

$$e^{3.1817} = 24.0974$$

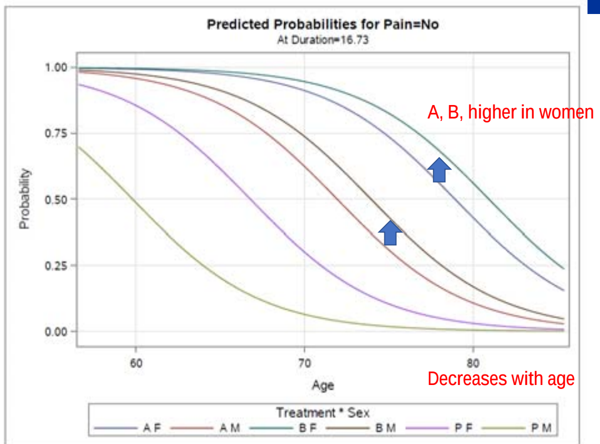
$$e^{3.7085} = 40.7942$$

Plot Prediction Probabilities

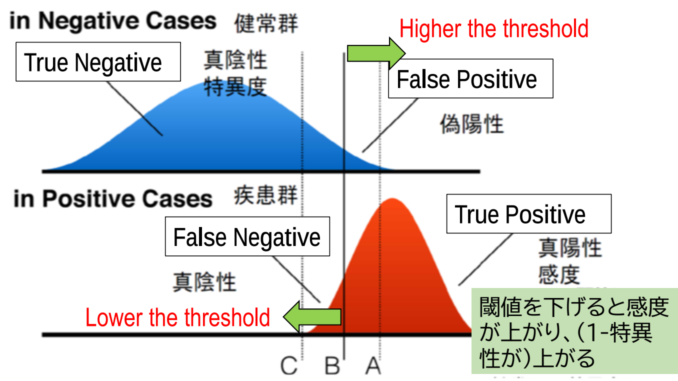
$$p = \frac{e^{a+bX}}{1+e^{a+bX}}$$

```
ods graphics on;
proc logistic data=Neuralgia PLOTS=(EFFECT);
class treatment sex / param=glm;
model pain= treatment sex age duration;
oddsratio Treatment;
run;
```

Plotting the predicted probability of extinction rate



ROC（Receiver Operating Characteristic）



感度・特異度

		Disease ⊕	Disease ⊖	Predictive Value	
Test	⊕	A True Positive (TP)	B False Positive (FP)	Positive Predictive Value (PPV) $\frac{TP}{TP+FP}=\frac{A}{A+B}$	Total Positive Results (A + B)
	⊖	C False Negative (FN)	D True Negative (TN)	Negative Predictive Value (NPV) $\frac{TN}{FN+TN}=\frac{D}{C+D}$	Total Negative Results (C + D)
Sensitivity & Specificity		Sensitivity $\frac{TP}{TP+FN}=\frac{A}{A+C}$	Specificity $\frac{TN}{FP+TN}=\frac{D}{B+D}$		
		All diseased patients (A + C)	All non-diseased patients (B + D)		

疾病を有する患者において
真陽性となる割合

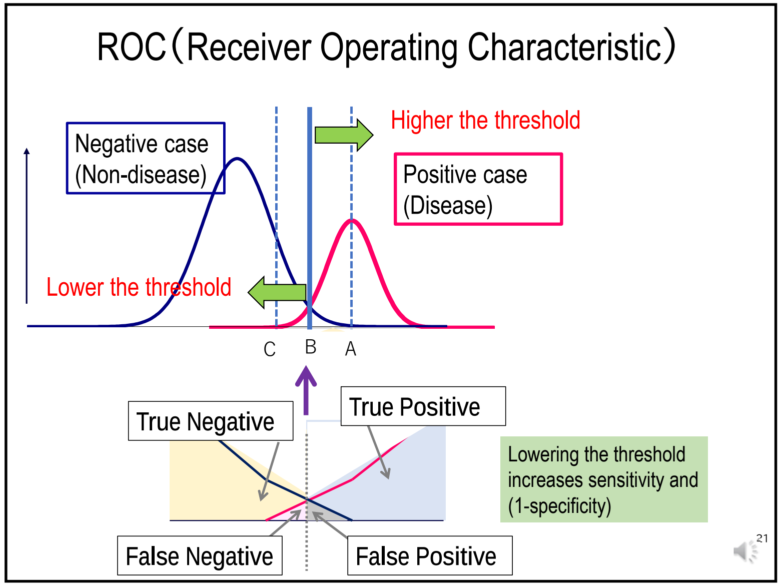
病気を持っていない患者さんの真
の陰性結果の割合を示しています。

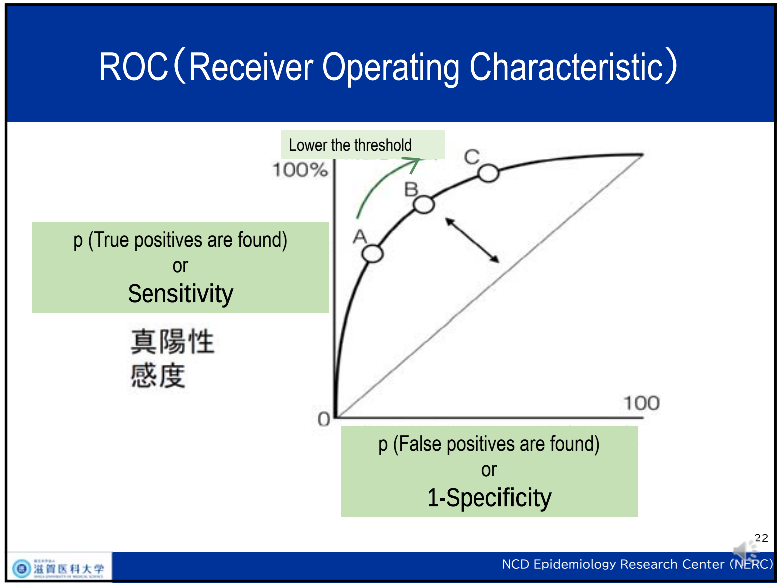
Positive (Negative) predictive value

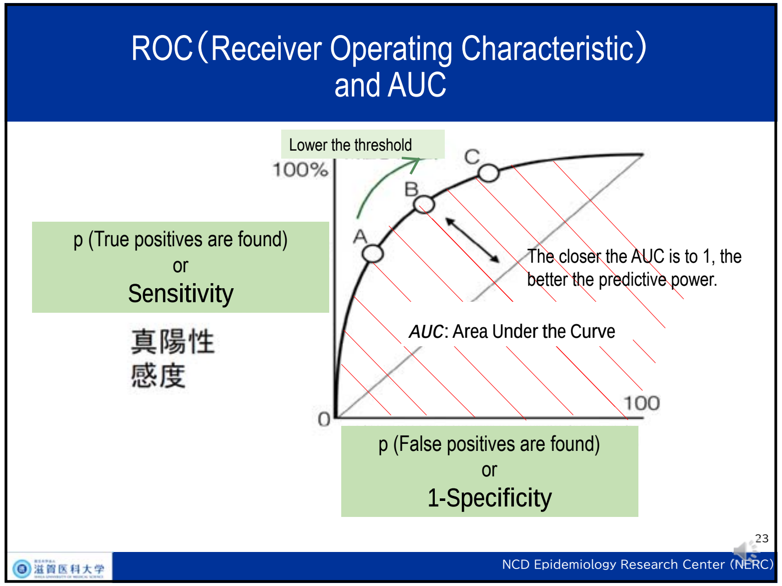
陽性の場合、
病気の存在を反
映する

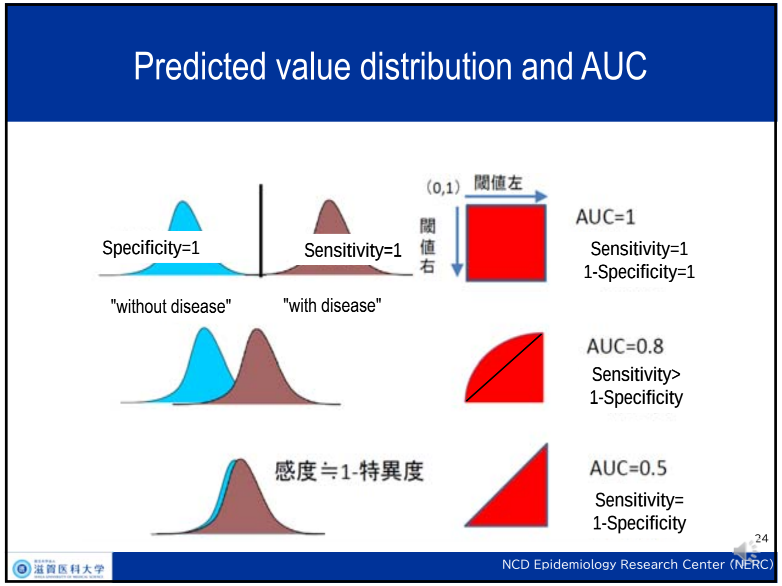
		Disease ⊕	Disease ⊖	Predictive Value	
Test	⊕	A True Positive (TP)	B False Positive (FP)	Positive Predictive Value (PPV) $\frac{TP}{TP+FP}=\frac{A}{A+B}$	Total Positive Results (A + B)
	⊖	C False Negative (FN)	D True Negative (TN)	Negative Predictive Value (NPV) $\frac{TN}{FN+TN}=\frac{D}{C+D}$	Total Negative Results (C + D)
Sensitivity & Specificity		Sensitivity $\frac{TP}{TP+FN}=\frac{A}{A+C}$	Specificity $\frac{TN}{FP+TN}=\frac{D}{B+D}$		
		All diseased patients (A + C)	All non-diseased patients (B + D)		

陰性は病気でないことを
意味します。


ROC（Receiver Operating Characteristic）
Negative case
(Non-disease)
Higher the threshold
Positive case
(Disease)
Lower the threshold
C
B
A
True Negative
True Positive
Lowering the threshold increases sensitivity and (1-specificity)
False Negative
False Positive
21


ROC（Receiver Operating Characteristic）
Lower the threshold
100%
C
B
A
p (True positives are found)
or
Sensitivity
真陽性
感度
0
100
p (False positives are found)
or
1-Specificity
22
NCD Epidemiology Research Center (NERC)


ROC（Receiver Operating Characteristic）
and AUC
Lower the threshold
100%
C
B
A
p (True positives are found)
or
Sensitivity
真陽性
感度
The closer the AUC is to 1, the better the predictive power.
AUC: Area Under the Curve
0
100
p (False positives are found)
or
1-Specificity
23
NCD Epidemiology Research Center (NERC)


Predicted value distribution and AUC
(0,1)
閾値左
閾値右
Specificity=1
Sensitivity=1
AUC=1
Sensitivity=1
1-Specificity=1
"without disease"
"with disease"
AUC=0.8
Sensitivity>
1-Specificity
感度≒1-特異度
AUC=0.5
Sensitivity=
1-Specificity
24
NCD Epidemiology Research Center (NERC)

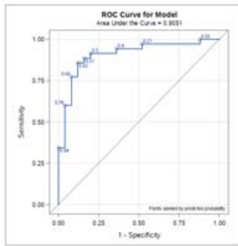
Create ROC curve and compare AUC Model data=Neuralgia

```
ods graphics on;
proc logistic data=Neuralgia
PLOTS=ROC(ID=PROB);
class treatment sex/param=glm;
model pain= treatment sex age duration;
roc 'age' age ;
roc 'duration' duration;
roccontrast reference(model)/estimate e;
run;
```

Compare with (1) this model, (2) Age only, and (3) Period only. model

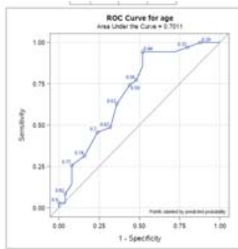
ROC and AUC in three models

(1) This model



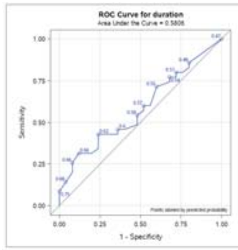
AUC=0.9051

(2) Age only model



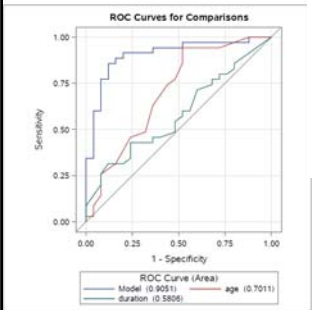
AUC=0.7011

(3) Duration only model



AUC=0.5806

ROC and AUC in three models



ROC Association Statistics

ROC Model	Mann-Whitney Area	Standard Error	95% Wald Confidence Limits		Somers' D	Gamma	Tau-a
Model	0.9051	0.0412	0.8244	0.9858	0.8103	0.8103	0.4006
age	0.7011	0.0717	0.5605	0.8418	0.4023	0.4211	0.1989
duration	0.5806	0.0744	0.4348	0.7263	0.1611	0.1693	0.0797

ROC Contrast Estimation and Testing Results by Row

Contrast	Estimate	Standard Error	95% Wald Confidence Limits		Chi-Square	Pr > ChiSq
age - Model	-0.2040	0.0743	-0.3496	-0.0584	7.5402	0.0060
duration - Model	-0.3246	0.0815	-0.4842	-0.1649	15.8789	<.0001

Is there a difference in AUC between the three models?

outcome variables (type and correlation) and analysis

Correlation of outcome variables

outcome variables	(−)	(+)
continuous	General linear model (GLM) ・regression analysis(REG) ・analysis of variance (ANOVA) ・t-test (TTEST)	General linear mixed model (MIXED)
binominal	Logistic regression (LOGISTIC)	Generalized linear mixed model (NLMIXED) (GLIMMIX)
multiple	Loglinear model (CATMOD)	
Count data	Poisson regression (GENMOD)	

Generalized linear model (GENMOD)

Poisson regression

- ポアソン回帰は、カウントデータや分割表をモデル化するために用いられる回帰分析の一般化線形モデル形式である。
- ポアソン回帰は, 応答変数Y がポアソン分布を持っていて, その期待値の対数が未知パラメータの線形結合によってモデル化できると仮定する.
- これは, 希少事象に適している.
  - 稀な事象
  - 保険金請求率
  - 機器の故障
  - 犯罪の発生

Poisson distribution

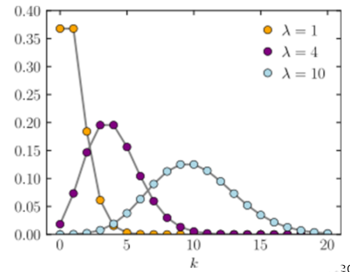
- 歪んでいる
- 非負の値のみをとる
- 分散は平均に等しい

$$f(k;\lambda) = \Pr(X{=}k) = \frac{\lambda^k e^{-\lambda}}{k!},$$

where

- k is the number of occurrences (k=0,1,2・・・)
- λ is equal to the expected value(E(x)) of X and also to its variance (V(E))

$$\lambda = \mathrm{E}(X) = \mathrm{Var}(X).$$



Poisson regression

$$\lambda_i = \exp(\beta_0 + \beta_1 x_i) = \exp\beta_0 \times \exp\beta_1 x_i$$

$$log(\lambda_i) = \beta_0 + \beta_1 x_i$$

1. ポアソン応答
応答変数は、ポアソン分布で記述される単位時間または単位空間あたりのカウントである。
2. 独立性
オブザベーションは互いに独立でなければならない。
3. 平均＝分散
定義上、ポアソン確率変数の平均は分散と等しくなければならない。
4. 線形性
平均率の対数、log(λ)は x

Example. Ear infection

- このデータは、ニューサウスウェールズ州水道局が1990年に行った「サーフ／ヘルス調査(Pilot Surf/Health Study)」によるものである。
- この研究の目的は、特に、ビーチで泳ぐ人が、ビーチで泳がない人よりも耳の感染症にかかるリスクが高いかどうかを判断することでした。
- データ集合は、5つの変数と287のオブザベーションを含むear infectionと名づけられました。

Variable	Variable Name	Variable Type	Variable Values
Frequent Ocean Swimmer	fos	Categorical	1=Yes 2=No
Location	location	Categorical	1=Non-beach 4=Beach
Age Group	agegroup	Ordinal	2=15 to 19 years 3=20 to 24 years 4=25 to 29 years
Sex	sex	Categorical	1=Male 2=Female
Number of Ear Infections	numearinfections	Continuous	0-17

Basic statistics
Freq procedure

swimmer	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Freq	143	49.83	143	49.83
Occas	144	50.17	287	100.00

location	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Beach	147	51.22	147	51.22
NonBeach	140	48.78	287	100.00

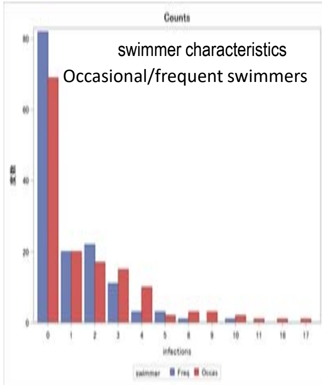
age	Frequency	Percent	Cumulative Frequency	Cumulative Percent
15-19	140	48.78	140	48.78
20-24	79	27.53	219	76.31
25-29	68	23.69	287	100.00

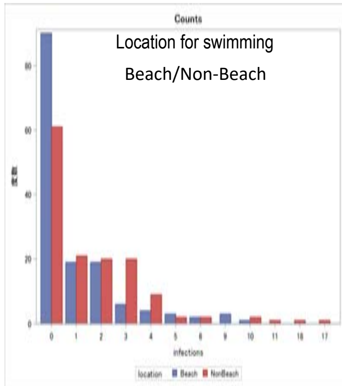
gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	99	34.49	99	34.49
Male	188	65.51	287	100.00

```
/*poisson earinfection*/
Data earinfection;
input swimmer $ location $ age $ gender $  infections;
datalines;
Occas     NonBeach15-19     Male      0
Occas     NonBeach15-19     Male      0
Occas     NonBeach15-19     Male      0
Occas     NonBeach15-19     Male      0
Occas     NonBeach15-19     Male      0
・
・   ====omission ====
・
Freq      Beach     25-29     Female    2
Freq      Beach     25-29     Female    2
;
run;

ods pdf file='F:¥pois1.pdf';
proc freq;
table swimmer location  age  gender  infections;
run;
ods pdf close;
```


Frequency of ear infections





Poisson analysis with
PROC GENMOD

PROC GENMOD DATA=*SAS-data-set <options>*;
CLASS *variables </option>*;
MODEL *response=predictors </options>*;
ESTIMATE *'label' effect values … <options>*;
OUTPUT OUT=*SAS-data-set keyword=name </option>*;
RUN;

Factors associated with the development of ear infections

```
PROC GENMOD data=earinfection;
CLASS swimmer (param=ref ref=first)
location (param=ref ref=first)
gender (param=ref ref=last);
age (param=ref ref=first);

MODEL infections = swimmer location gender age
/ dist=poi link=log type3;
RUN;
```

Variable	Variable Name	Variable Type	Variable Values
Frequent Ocean Swimmer	fos	Categorical	1=Yes 2=No
Location	location	Categorical	1=Non-beach 4=Beach
Age Group	agegroup	Ordinal	2=15 to 19 years 3=20 to 24 years 4=25 to 29 years
Sex	sex	Categorical	1=Male 2=Female
Number of Ear Infections	numearinfections	Continuous	0-17

Factors associated with the development of ear infections

```
PROC GENMOD data=earinfection;
CLASS swimmer (param=ref ref=first)
location (param=ref ref=first)
gender (param=ref ref=last)
age (param=ref ref=first);
MODEL infections = swimmer location age gender/ dist=poi link=log type3;
ESTIMATE 'Occasional vs. Frequent swimmer'
swimmer 1 / exp;
ESTIMATE 'Non-Beach vs. Beach' location 1 / exp;
ESTIMATE 'Female vs. Male' gender 1 / exp;
ESTIMATE ' 20-24 vs. 15-19 ' age 1 -1 / exp;
ESTIMATE ' 25-29 vs. 15-19 ' age 0 -1 / exp;
ESTIMATE ' 20-24 vs. 25-29 ' age 1  0 / exp;
RUN;
```

To calculate IRR, calculate the value of y and exp from log(y).

Result of Genmod

The GENMOD Procedure

Model Information	
Data Set	WORK.EARINFECTION
Distribution	Poisson
Link Function	Log
Dependent Variable	infections

Number of Observations Read	287
Number of Observations Used	287

Class Level Information

Class	Value	Design Variables	
swimmer	Freq	0	
	Occas	1	
location	Beach	0	
	NonBeach	1	
gender	Female	1	
	Male	0	
age	15-19	0	0
	20-24	1	0
	25-29	0	1

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	281	755.4332	2.6884
Scaled Deviance	281	755.4332	2.6884
Pearson Chi-Square	281	949.7387	3.3799
Scaled Pearson X2	281	949.7387	3.3799
Log Likelihood		-233.3284	
Full Log Likelihood		-563.9140	
AIC (smaller is better)		1139.8280	
AICC (smaller is better)		1140.1280	
BIC (smaller is better)		1161.7849	

The theoretical value is 1; if greater than 1, Over-dispersion

Result of Genmod
Poisson regression

Analysis Of Maximum Likelihood Parameter Estimates

Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	-0.2125	0.1230	-0.4535	0.0286	2.98	0.0841
swimmer	Occas	1	0.6115	0.1050	0.4057	0.8173	33.91	<.0001
location	NonBeach	1	0.5345	0.1067	0.3254	0.7436	25.11	<.0001
age	20-24	1	-0.3744	0.1284	-0.6260	-0.1228	8.51	0.0035
age	25-29	1	-0.1897	0.1301	-0.4447	0.0653	2.13	0.1447
gender	Female	1	0.0899	0.1123	-0.1303	0.3100	0.64	0.4237
Scale		0	1.0000	0.0000	1.0000	1.0000		

Contrast Estimate Results

label	Mean Estimate	Mean Confidence Limits		L'Beta estimate	Standard Error	α	L'Beta Confidence Limits		Chi-Square	Pr > ChiSq
Occasional vs. Frequent swimmer	1.8432	1.5003	2.2644	0.6115	0.1050	0.05	0.4057	0.8173	33.91	<.0001
Exp(Occasional vs. Frequent swimmer)				**1.8432**	0.1935	0.05	1.5003	2.2644		
Non-Beach vs. Beach	1.7067	1.3846	2.1036	0.5345	0.1067	0.05	0.3254	0.7436	25.11	<.0001
Exp(Non-Beach vs. Beach)				**1.7067**	0.1821	0.05	1.3846	2.1036		
Female vs. Male	1.0940	0.8779	1.3634	0.0899	0.1123	0.05	-0.1303	0.3100	0.64	0.4237
Exp(Female vs. Male)				1.0940	0.1229	0.05	0.8779	1.3634		
20-24 vs. 15-19	1.2029	0.8871	1.6310	0.1847	0.1554	0.05	-0.1198	0.4892	1.41	0.2346
Exp(20-24 vs. 15-19)				1.2029	0.1869	0.05	0.8871	1.6310		
25-29 vs. 15-19	1.4542	1.1307	1.8701	0.3744	0.1284	0.05	0.1228	0.6260	8.51	0.0035
Exp(25-29 vs. 15-19)				**1.4542**	0.1867	0.05	1.1307	1.8701		
25-29 vs. 20-24	0.8272	0.6410	1.0674	-0.1897	0.1301	0.05	-0.4447	0.0653	2.13	0.1447
Exp(25-29 vs. 20-24)				0.8272	0.1076	0.05	0.6410	1.0674		

- Frequency of swimming, location, and age (25-29 years) were significant factors for ear inflammation

Over-dispersion

- 結果変数の分散が期待値より大きい
- ポアソン分布の分散の理論値は、期待値と等しい。
- 過大分散は、標準誤差の過小評価やカイ二乗統計量の過大評価につながることがある。

- カウントデータの分散は平均値より大きいことが多い 過分散
- なぜ過大分散が起こるのか？
- モデルに含まれていない重要な効果がある場合、平均値の差を十分に説明できない。
- 外れ値が存在する。
- 変数間に正の相関がある

adjust for over-dispersion

```
PROC GENMOD data=earinfections;
CLASS swimmer (param=ref ref=first)
location (param=ref ref=first)
gender (param=ref ref=last)
age (param=ref ref=first);
MODEL infections = swimmer location gender age
/ dist=poi link=log type3 pscale;
RUN;
```


Result after adjustment ①

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	281	755.4332	2.6884
Scaled Deviance	281	223.5107	0.7954
Pearson Chi-Square	281	949.7387	3.3799
Scaled Pearson X2	281	281.0000	1.0000
Log Likelihood		-69.0351	
Full Log Likelihood		-563.9140	
AIC (smaller is better)		1139.8280	
AICC (smaller is better)		1140.1280	
BIC (smaller is better)		1161.7849	

Result after adjustment ②

Analysis Of Maximum Likelihood Parameter Estimates

Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	-0.2125	0.2261	-0.6557	0.2308	0.88	0.3475
swimmer	Occas	1	0.6115	0.1930	0.2331	0.9898	10.03	0.0015
location	NonBeach	1	0.5345	0.1961	0.1501	0.9189	7.43	0.0064
gender	Female	1	0.0899	0.2065	-0.3148	0.4945	0.19	0.6635
age	20-24	1	-0.3744	0.2360	-0.8370	0.0881	2.52	0.1126
age	25-29	1	-0.1897	0.2392	-0.6585	0.2790	0.63	0.4276
Scale		0	1.8384	0.0000	1.8384	1.8384		

LR Statistics For Type 3 Analysis

Source	Num DF	Den DF	F Value	Pr > F	Chi-Square	Pr > ChiSq
swimmer	1	281	10.51	0.0013	10.51	0.0012
location	1	281	7.62	0.0062	7.62	0.0058
gender	1	281	0.19	0.6651	0.19	0.6648
age	2	281	1.36	0.2575	2.73	0.2558

Result after adjustment ③

Contrast Estimate Results

Label	Mean Estimate	Mean Confidence Limits		L'Beta Estimate	Standard Error	Alpha	L'Beta Confidence Limits		Chi-Square	Pr > Chi Sq
Occasional vs. Frequent swimmer	1.8432	1.2625	2.6908	0.6115	0.1930	0.05	0.2331	0.9898	10.03	0.0015
Exp(Occasional vs. Frequent swimmer)				1.8432	0.3558	0.05	1.2625	2.6908		
Non-Beach vs. Beach	1.7067	1.1620	2.5066	0.5345	0.1961	0.05	0.1501	0.9189	7.43	0.0064
Exp(Non-Beach vs. Beach)				1.7067	0.3347	0.05	1.1620	2.5066		
Female vs. Male	1.0940	0.7299	1.6397	0.0899	0.2065	0.05	-0.3148	0.4945	0.19	0.6635
Exp(Female vs. Male)				1.0940	0.2259	0.05	0.7299	1.6397		
20-24 vs. 15-19	1.2029	0.6872	2.1055	0.1847	0.2856	0.05	-0.3752	0.7445	0.42	0.5179
Exp(20-24 vs. 15-19)				1.2029	0.3436	0.05	0.6872	2.1055		
25-29 vs. 15-19	1.4542	0.9157	2.3093	0.3744	0.2360	0.05	-0.0881	0.8370	2.52	0.1126
Exp(25-29 vs. 15-19)				1.4542	0.3432	0.05	0.9157	2.3093		
25-29 vs. 20-24	0.8272	0.5176	1.3219	-0.1897	0.2392	0.05	-0.6585	0.2790	0.63	0.4276
Exp(25-29 vs. 20-24)				0.8272	0.1978	0.05	0.5176	1.3219		

- 水泳の頻度、場所、年齢(25-29歳)が耳の炎症の有意な要因であった

Offset

- サンプル出現率はY(イベント数)/t(時間または空間)です。
- この割合の期待値は
- E(Y /t) = 1/t × E(Y) = μ/t
- 事象の発生率の期待値に対するポアソン回帰モデルは
- log(μ/t) = α + βx
- log(μ) - log(t) = α + βx
- log(μ) = α + βx + log(t)

$$\log(\mu/t) = \alpha + \beta x$$
$$\mu/t = \exp^{\alpha}\exp^{\beta x}$$
$$\mu = t\exp^{\alpha}\exp^{\beta x}$$

- log(t)」は調整項であり、個人によってtの値が異なる場合がある。
- - log(t)は「オフセット」と呼ばれる。

No index of fractions is used

- 単位あたりのカウント数に興味がある場合、応答変数としてパーセンテージを使用する傾向がある
- ポアソン分布に従わなくなるだけでなく、分子のカウント数、分母のユニット数の情報も失われます。
- 例

5 /10events, 500 /1000events は同じ扱いになる

Report

本講義のレポート課題である。

Q1 ロジスティック回帰の講義で使用した「神経痛の鎮痛剤に関する研究」のデータについて、再度解析を再現し、その出力と結果の読解を確認する。

Q2 ポアソン回帰の講義で使用した「耳の感染症」の例について、再度解析を再現し、その出力と結果の読み方を確認しなさい。